

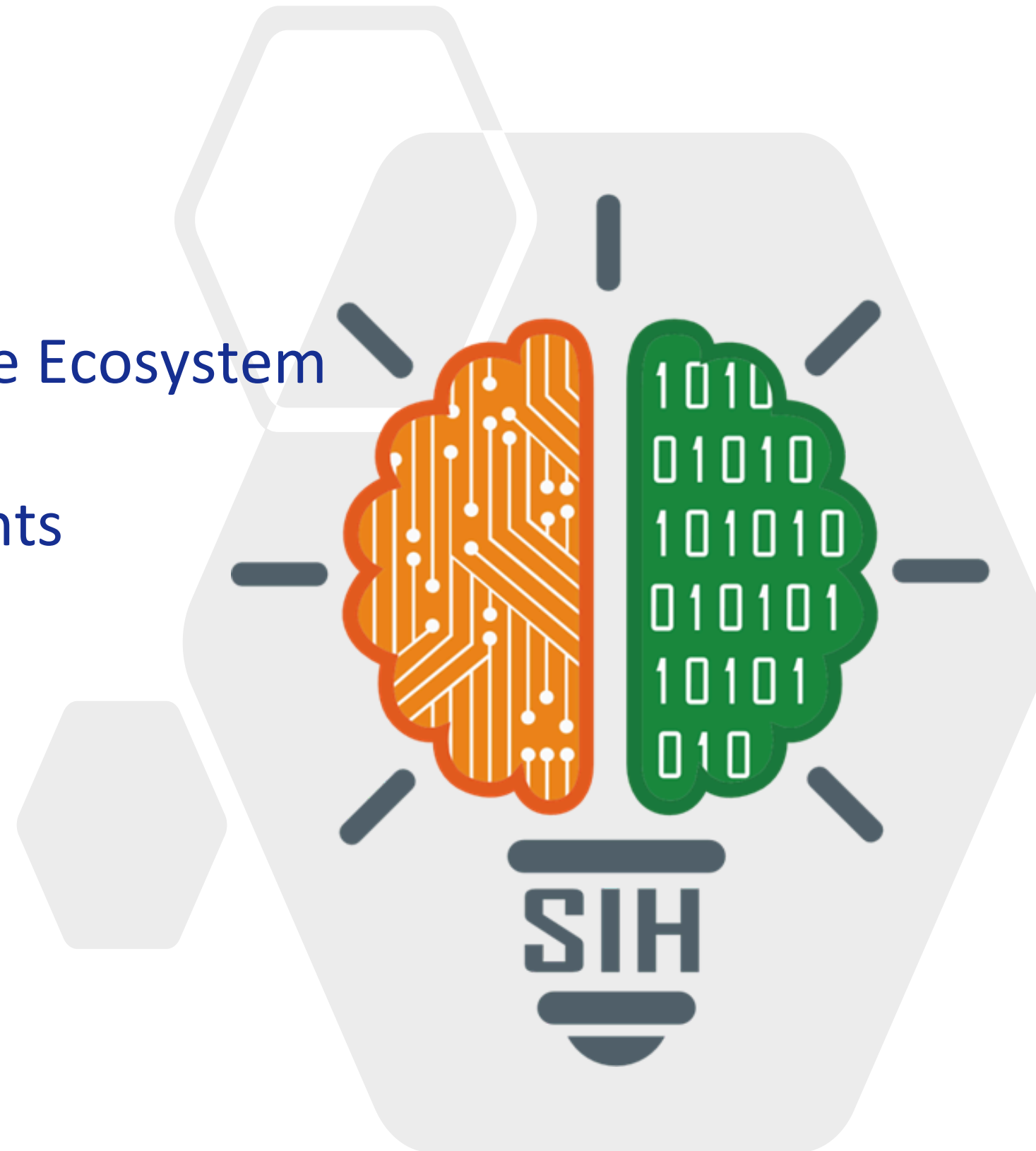
SMART INDIA HACKATHON 2026



- **Problem Statement ID –** SIH26176
- **Problem Statement Title-** ORCA Marine Ecosystem

Reasoning with Collaborative Agents

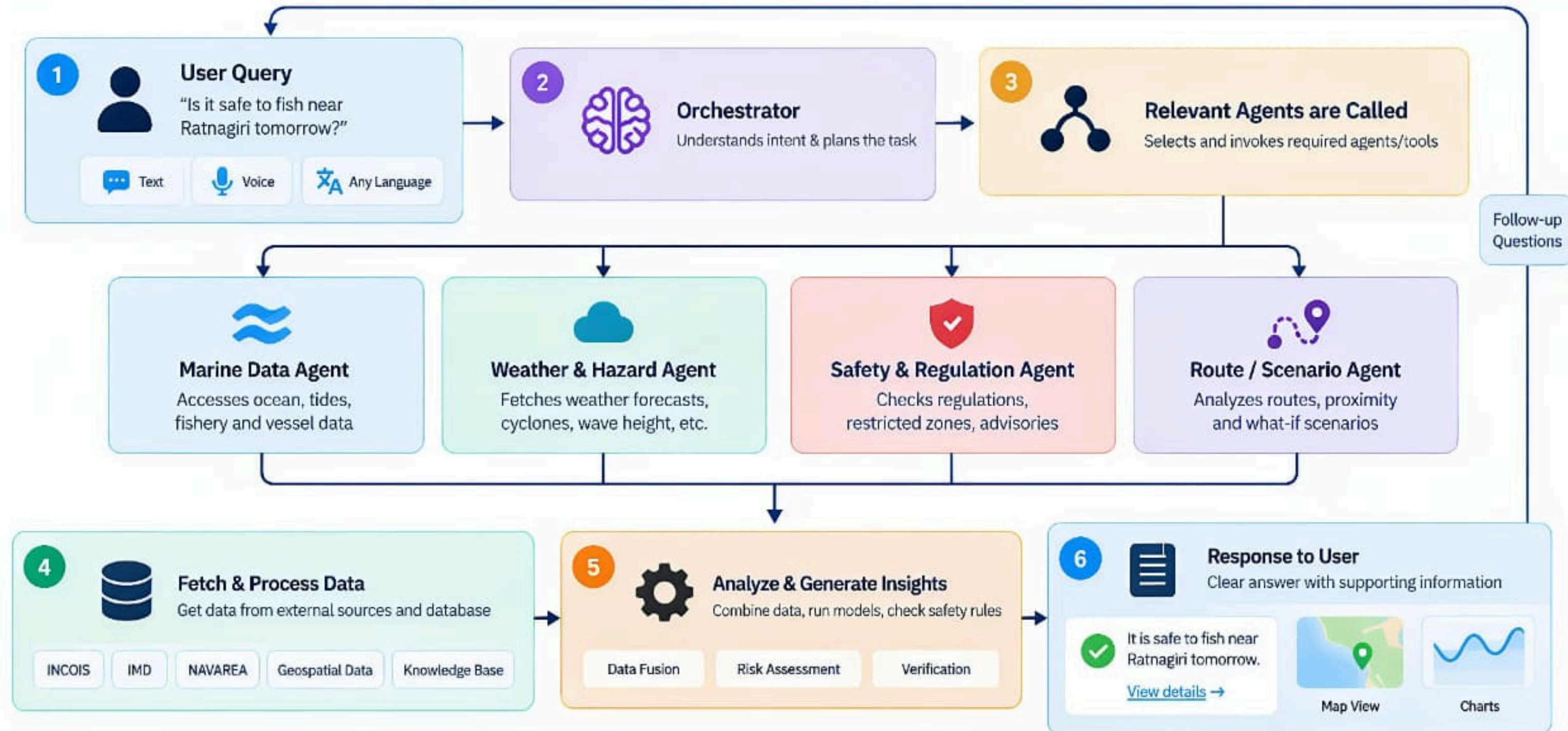
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team Name:** CodeCatalysts
- **Team ID :-**



Five real problems. One platform that solves them.

PROBLEMS FACED BY USERS		OUR SOLUTION (ORCA)	INNOVATION FACTOR
01	 Fragmented Information Data scattered across sources.	 Unified Marine Data Integration Satellite, ocean, weather, GIS in one.	 Multi-Source Intelligence Multiple data sources, analyzed together.
02	 Difficult Interpretation Complex data, hard to understand.	 Multilingual Conversational AI Simple, actionable insights.	 Marine-Specific Reasoning Understands spatial and temporal patterns.
03	 Limited Personalized Decision Support Generic information.	 Context-Aware Recommendations Insights based on real-time context.	 Route Planning and Geofencing Recommends safe routes and restricted areas.
04	 Safety, Restrictions & Alerts Timely alerts are hard to access.	 Integrated Safety, Restrictions & Alerts Real-time warnings and advisories.	 Voice-Enabled Interaction Hands-free, natural voice queries.
05	 Lack of Transparency Unclear why a recommendation is given.	 Explainable & Evidence-Based Decisions Shows sources and key reasoning.	 Multilingual Support Answers in multiple Indian languages with traceable data.

TECHNICAL APPROACH



FRONTEND

User Interface & Visualization



React



Next.js



TypeScript



Google Maps



BACKEND

APIs & Business Logic



FastAPI



Python



AI / ML LAYER

Language Models



LLM

(OpenRouter, Sarvam)



Our Models



DATA LAYER

Storage & Knowledge



PostgreSQL

Structured data storage



PostGIS

Spatial & geospatial data



DEVELOPMENT & DEVOPS

Hosting, Scaling & Operations



Docker

Containerization

Why ORCA Works

Feasible to build. Built to scale. Priced to matter.



FEASIBILITY

Can we build it? Yes.



Data

Uses existing marine datasets and open APIs.



Operational

Automates fragmented marine information.



Scalable

Modular multi-agent architecture enables incremental growth.



SCALABILITY

Built to grow beyond the prototype.



More AI Agents

Add domain agents easily (e.g., pollution, fisheries, disaster).



More Users

Fishermen, researchers, government, ports and more.



Modular Agents

MCP-based architecture for easy expansion.



UNIQUE SELLING POINT

Real impact, wider adoption.



Social Impact

Safer navigation and sustainable fisheries for coastal communities.



Market Potential

Applicable across fisheries, shipping, ports and disaster management.



Adoption Potential

Simple conversational interface enables non-technical users.

IMPACT & BENEFITS

Real-world value through intelligent, safe and trusted marine decision intelligence.

KEY BENEFITS

WHAT DOES ORCA PROVIDE?



UNIFIED INTELLIGENCE

All marine data in one platform.



Reduces fragmented information access.



FASTER DECISIONS

Ask → Analyze → Answer instantly.



Reduces manual effort and time.

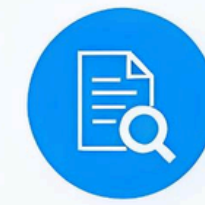


SAFER DECISIONS

Safety built into every recommendation.



Prevents risks and supports compliance.



TRUSTED AI OUTPUT

Evidence-backed, verifiable answers.



Transparent and explainable decisions.



MULTILINGUAL ACCESS

Insights in your own language.



Makes marine information accessible to all users.

KEY IMPACT

WHAT LARGER CHANGE CAN IT CREATE?



ENHANCED MARITIME SAFETY

Better awareness of hazards.



Fewer accidents, safer communities.



SMARTER FISHERIES

Data-backed fishing potential and advisories.



Improved catch, higher livelihoods.



DATA-DRIVEN MANAGEMENT

Real-time + historical spatial intelligence.



Better policy and resource planning.



EFFICIENT ROUTE PLANNING

Alerts on restricted or hazardous areas.



Safer and lower cost operations.



SUSTAINABLE OCEAN FUTURE

Supports conservation and climate resilience.



Healthier oceans for future generations.

Research & References

Built on trusted sources. Focused on real-world impact.

01

Research Foundation

Proven approaches we build on



FloatChat (Ecological Informatics, 2026)

Unified RAG for ocean data
(PostgreSQL + FAISS + RAG).



Strong results

86% queries resolved;
94% valid SQL generation.



Our extension

Multi-source marine decision
support (weather + hazards +
fishing zones + routing).

02

Indian Government Data

Authoritative & relevant for SIH



INCOIS

Ocean forecasts, PFZ, currents,
SST, cyclones.



IMD

Marine warnings, coastal bulletins,
cyclone tracks, fishermen advisories.



ISRO / NRSC

Ocean colour, SST, disaster
management support.



Trusted, open datasets

with timestamps and spatial coverage.

03

ORCA Data Stack

Integrated, scalable, open



Ocean & weather data

INCOIS OSF / PFZ, Copernicus Marine,
IMD, Open-Meteo.



Earth observation

ISRO/EOS ocean-colour, SST,
chlorophyll.



Geospatial

PostGIS + official boundaries
(Marine Regions / Protected Planet).



Frontend & maps

MapLibre GL + real-time layers
(routes, hazards, evidence overlays).